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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

USE OF REMIFENTANIL PCA FOR LABOUR ANALGESIA

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1. Introduction and purpose

Remifentanil is a safe and effective option for labour analgesia. Acting within 1 - 2 minutes and suited to patient controlled administration, it offers another analgesic option for women in labour. For safe use, the parturient will require careful and continuous monitoring by the midwife during labour.

2. Scope

This guideline is intended as reference for trainee anaesthetists and midwives who wish to provide patient controlled remifentanil analgesia for parturients in labour.

3. Information and safety

3.1 Indications

- If epidural analgesia for labour is contraindicated
- As an alternative to intramuscular (IM) diamorphine for parturients who do not want an epidural
- Labour analgesia following intrauterine death
- Labour analgesia during siting of an epidural
- Analgesia during the repair of tears and episiotomies

3.2 Criteria for use

- Parturient must be over 36 weeks gestation and in active labour
- More than 4 hours must have elapsed since the last dose of IM diamorphine (if administered)
- The Midwifery Shift Leader has confirmed that there is appropriate staffing on labour ward to enable the parturient to have **continuous** trained midwifery presence in the labour room

3.3 Consent

- The parturient will need to have read and understood the patient information sheet given to her by the midwife

- Explain that Remifentanil PCA can be used in conjunction with Entonox

- The anaesthetist should inform the parturient of possible side-effects including drowsiness, itch, nausea and dizziness. In particular that approximately 1 in 10 of parturients using a remifentanil PCA in labour will experience transient lowered SpO_2 levels requiring the administration of oxygen via nasal specs, and that 1 in 100 require the discontinuation of the PCA.

- The parturient should be shown how to use the PCA and should be told to press the button just before or at the start of a contraction.

- Explain that **only** the parturient herself should use the PCA button

3.4 Setting up the PCA

- The anaesthetist on call for the delivery suite is responsible for prescribing and setting up the remifentanil PCA along with the replacing empty pump syringes, while the midwife is responsible for monitoring the patient, the PCA device and ensuring that only the parturient presses the PCA button. A midwife information sheet is also available for their reference.

- Use a dedicated Arcomed PCA device (kept in labour ward store room, with key for pump attached to the 551 pager)

- Use the Remifentanil PCA Anaesthetic Checklist (attached to the front of the midwife information sheet and observation chart, which is kept in the Remifentanil PCA drawer on labour ward)

- ODP to issue 2mg of remifentanil from the obstetric theatre CD cupboard

- Dilute 2mg of remifentanil in a 50ml syringe of normal saline to make up a 40 mcg/ml solution

- Label the syringe clearly with the drug name, dilutant and concentration

- Connect the extension tubing onto the syringe, and prime the line

- Load the syringe into the PCA pump

- Ensure the remifentanil profile is selected on the pump (select ‘Maternity’ – 5511, then ‘Remifentanil’ – 1115) and check that the bolus dose (40mcg / 1ml) and lockout (2 minutes) are correct

- Connect the PCA to a **dedicated** cannula (22g (blue cannula) is sufficient). Remove this cannula **without flushing** when the PCA is no longer needed.

- Ensure that the BP cuff is on the opposite arm to the intravenous cannula used for the PCA, and that no other fluids are run through the same intravenous cannula as the PCA

- Prescribe the PCA on the patient's prescription chart (as well as Naloxone). The PCA prescription should read:

  Remifentanil PCA (2mg in 50 mls)
  Bolus dose : 40 mcg (1ml)
  Lockout period : 2 minutes

- Dilute 400mcg naloxone in a labelled 10ml syringe of normal saline (40mcg/ml). This is to be kept by the midwife in the room

- Ensure the oxygen saturation probe has been attached to patient, and that both patient and midwife are happy

- The Remifentanil PCA Anaesthetic Checklist should then be completed

- Start the remifentanil PCA and leave the labour room

- Enter the patient and procedure details in the Obstetric Anaesthesia Procedure and Follow up Book

- The anaesthetist **must remain on labour ward and be immediately available for the first 30 minutes** after starting the PCA. **If the anaesthetist is unable to, the PCA must be stopped**

3.5 Monitoring and Equipment

- **Continuous presence of a trained midwife in the labour room with the parturient at all times** while the remifentanil PCA is being used. If staffing levels or labour ward acuity means this is not possible, **then we cannot offer a remifentanil PCA**.

- **Continuous oxygen saturation monitoring**

- **A bag valve mask should always be available in the room**

- CTG monitoring is not required while the PCA is in progress unless otherwise indicated

- **Documentation of the following observations on the Remifentanil PCA Observation Chart** : Oxygen saturations, respiratory rate, sedation score and pain score, which should be documented every 30 minutes on the Remifentanil PCA Observation Chart.

- **Do not give any other drugs or fluids via the PCA cannula**

- **If oxygen saturations fall < 90%, oxygen via nasal cannula should be commenced at 2L/minute** (1 in 10 women suffer significant respiratory depression with remifentanil). Oxygen saturations < 90% despite nasal oxygen happens in fewer than 1% of women and will require discontinuation of the PCA

3.6 Indications for contacting the anaesthetist

- If the midwife has any concerns
- Sedation score of P or below (AVPU)
- Respiratory rate below 8/min
- Oxygen saturation below 90% despite 2L/min nasal oxygen.

IN THIS SITUATION THE MIDWIFE SHOULD IMMEDIATELY:

- **Stop the remifentanil PCA**
- **Call the anaesthetist to urgently attend**
- **Give an 80mcg (2ml bolus of the 40mcg/ml diluted naloxone) every 1 minute until the respiratory rate is > 8/min**

4. References

1. Volikas I, Butwick A, Wilkinson C. Maternal and neonatal side-effects of remifentanil patient-controlled analgesia in labour. BJA 2005; 95: 504